

# MULTI-WINNER VOTING: REPRESENTATION, MANIPULATION, DELIBERATION

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PhD Defense of **Oliviero Nardi**

DBAI, TU Wien – 16 October, 2026

Available at [tinyurl.com/oliviero-defense](https://tinyurl.com/oliviero-defense)

Multi-Winner Voting:  
Representation, Manipulation, Deliberation

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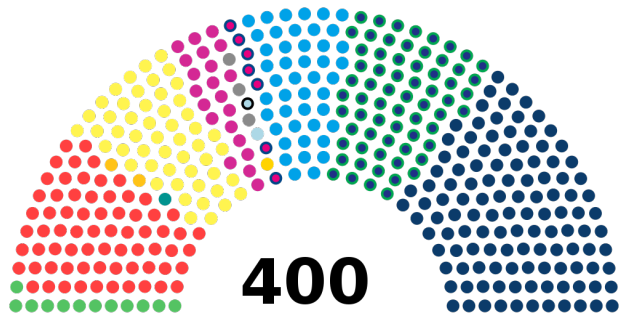
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Representation, Manipulation, Deliberation

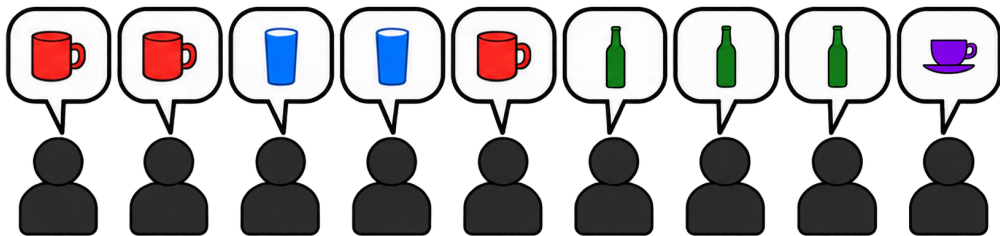
## Multi-Winner Voting:

Representation, Manipulation, Deliberation









**MajoritarianMango**

In systems with PR, opinions are too strongly divided and nothing gets done.

**ProportionalParrot**

A party's share of seats in the House of Commons should reflect its share of the popular vote.

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**PastTheP**

FPTP results

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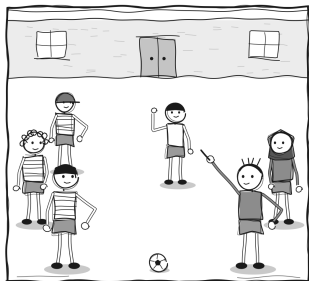
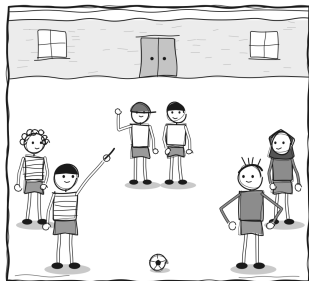
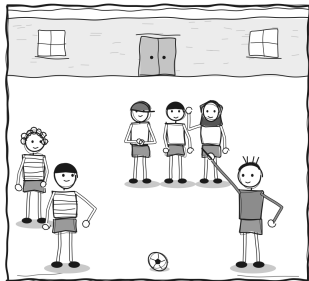
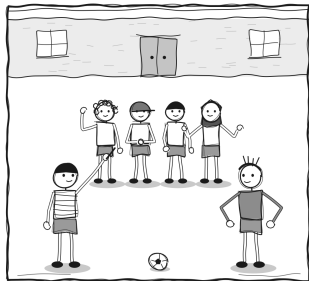
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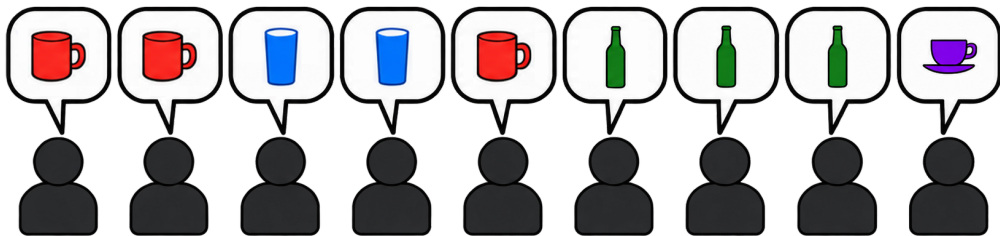


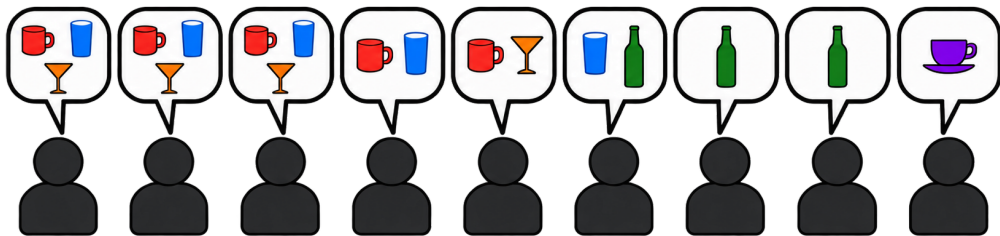
## Multi-Winner Voting:

Representation, Manipulation, Deliberation

(Approval-Based)  
Multi-Winner Voting:

Representation, Manipulation, Deliberation





**Vote for any number  
of options.**

☐

Joe Smith

☒

John Citizen

☐

Jane Doe

☐

Fred Rubble

☒

Mary Hill

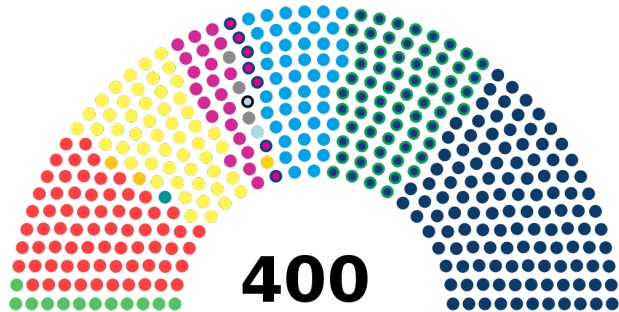
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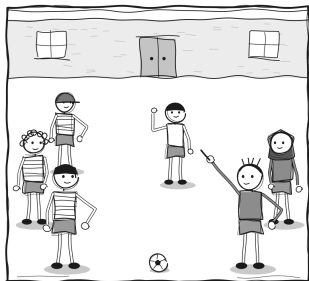
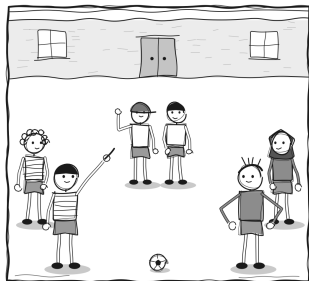
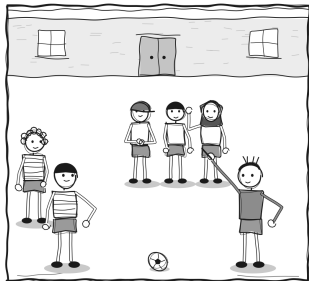
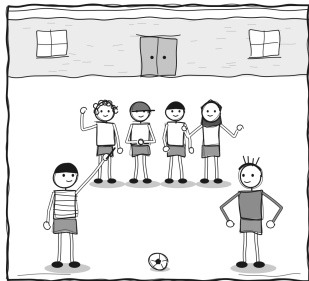
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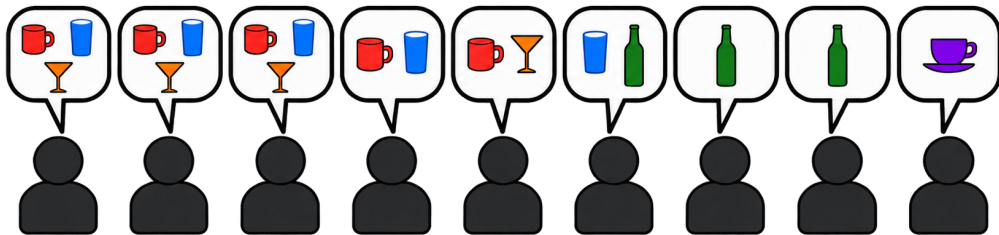
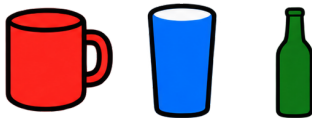
Multi-Winner Voting:

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## 1. REPRESENTATION



## COMMITTEE VOTING, INFORMALLY



## COMMITTEE VOTING, FORMALLY

- Voters  $N = \{1, 2, \dots, n\}$
- Alternatives  $C = \{a, b, \dots\}$
- Target committee size  $k \in \mathbb{N}$
- Each voter  $i \in N$  submits an approval ballot  $A_i \subseteq C$
- Collectively, we have a profile of preferences  $\mathbf{A} = (A_1, A_2, \dots, A_n)$
- A voting rule  $F$  maps each profile  $\mathbf{A}$  to a winning coalition  $W \subseteq C$  with  $|W| = k$

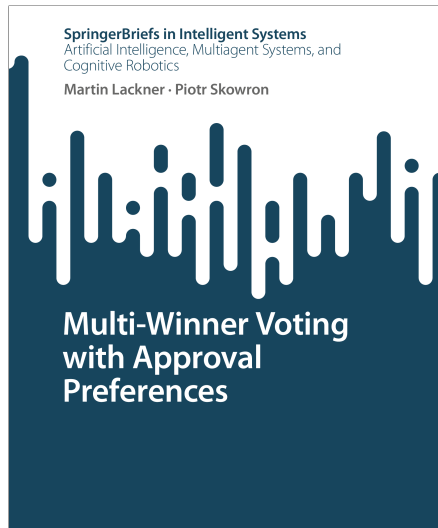
Additionally, we define  $N(c) = \{i \in N : c \in A_i\}$ .

## THE STANDARD CONCEPTION: AVOIDING UNDERREPRESENTATION

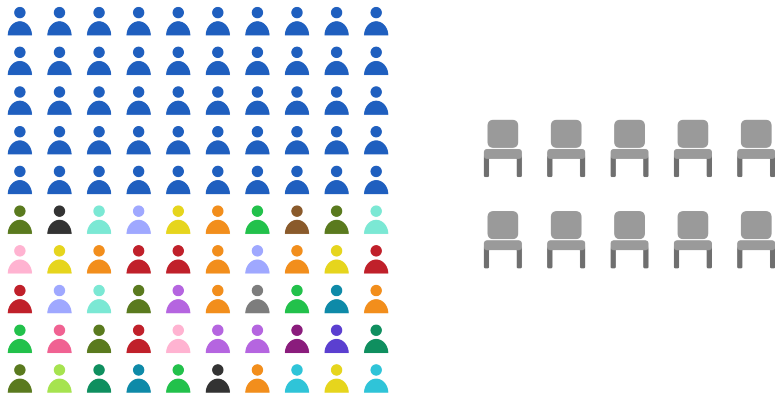
‘if a cohesive group of voters makes up  $x\%$  of the population,  
then it should get at least  $x\%$  of the seats’

# AVOIDING UNDERREPRESENTATION: STATE OF THE ART

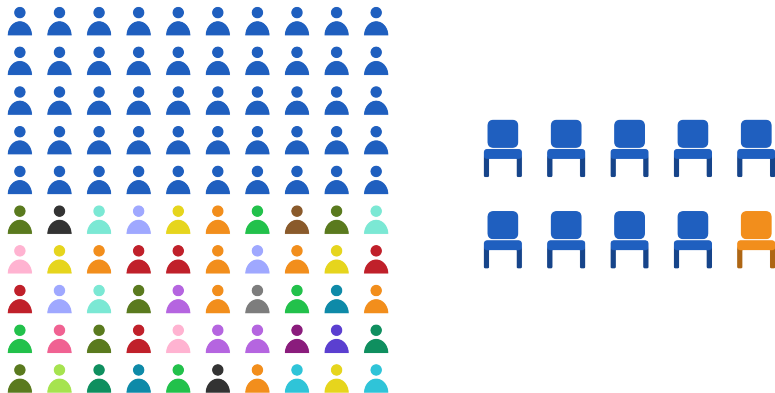
- Well-understood and formalised
  - Extended Justified Representation
  - Priceability
  - ...
- Natural and appealing rules provably satisfy it
  - Proportional Approval Voting
  - Method of Equal Shares
  - ...
- **We argue:** Not the last word on representation!



# A PROBLEM WITH ‘REPRESENTATIVE’ RULES



# A PROBLEM WITH ‘REPRESENTATIVE’ RULES



## ANOTHER CONCEPTION: AVOIDING OVERREPRESENTATION

‘if a cohesive group of voters makes up  $x\%$  of the population,  
then it should **not** decide more than  $x\%$  of the seats’

## WHY AVOIDING OVERREPRESENTATION?

- Fairness:
  - Prevents groups from having ‘too much’ power
  - Avoids ‘envy’
- Might prevent strategic ‘bullet voting’
- Technical applications
  - Safety in blockchain technologies

## FORMALISING OVERREPRESENTATION

‘if a cohesive group of voters makes up  $x\%$  of the population,  
then it should not decide more than  $x\%$  of the seats’

## OVERREPRESENTATION: TENTATIVE DEFINITION

In a winning coalition  $W$ , for every  $c \in W$ ,  
the group of voters  $N(c)$  should not have

$$|A_i \cap W| > \left\lceil k \frac{|N(c)|}{n} \right\rceil \text{ for all } i \in N(c).$$

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## THE DEFINITION HAS A BUG

Say we want to select  $k = 6$  candidates. Consider the following profile:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Each voter can get at most  $\lceil 6 \cdot \frac{1}{3} \rceil = 2$  candidates. Any committee overrepresents some voter:

- $\{a, b, c, d, e, f\}$  overrepresents voter 1 and voter 2
- $\{a, b, c, d, e, g\}$  overrepresents voter 1
- etc...

## JUSTIFIED UPPER QUOTA (JUQ)

In a winning coalition  $W$ , for every  $c \in W$ , if the group of voters  $N(c)$  has

$$|A_i \cap W| > \left\lceil k \frac{|N(c)|}{n} \right\rceil \quad \text{for all } i \in N(c),$$

then there cannot be  $d \in C \setminus W$  and  $N' \subseteq N(d)$  such that

$$|A_i \cap W \setminus \{c\} \cup \{d\}| \leq \left\lceil k \frac{|N'|}{n} \right\rceil \quad \text{for all } i \in N'.$$

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## JUQ FIXES THE BUG...

Let  $k = 6$  and consider again:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Recall: each voter can get at most  $\lceil 6 \cdot \frac{1}{3} \rceil = 2$  candidates.

Committee  $\{a, b, c, d, e, g\}$  **satisfies** JUQ:

- **Voter 1** is overrepresented, must kick out  $c$
- But: **voter 2** already at quota
- Thus, no need to swap  $c$  with  $f$

## ...BUT NOT TRIVIALY

Let  $k = 6$  and consider again:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Recall: each voter can get at most  $\lceil 6 \cdot \frac{1}{3} \rceil = 2$  candidates.

Committee  $\{a, b, c, d, e, f\}$  **fails** JUQ:

- **Voter 1** is overrepresented, must kick out  $c$
- **Voter 3** is underrepresented
- Thus, must swap  $c$  with  $g$

## WHICH RULES SATISFY JUQ?

Consider the standard family of Thiele rules:

### Definition

Given a non-increasing  $\omega: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ , the  $\omega$ -Thiele rule picks a committee maximising

$$\sum_{i \in N} \sum_{\ell=1}^{|A_i \cap W|} \omega(\ell)$$

Terrifying news:

### Proposition

*No Thiele rule satisfies JUQ.*

## EXTENDING THIELE RULES

We define **composite Thiele rules**, which work in stages of tiebreaking:

### Definition

Given a **sequence** of non-increasing  $\omega_1, \omega_2, \dots: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ , the corresponding composite Thiele rule picks a committee as follows:

1. First, select **all** committees maximising the score of  $\omega_1$ ;
2. Among those, select all committees maximising the score of  $\omega_2$ ;
3. Repeat until you are left with a single committee (or without scoring functions).

### Theorem

*There is a **unique** composite Thiele rule that satisfies JUQ: we called it **Adams Approval Voting**.*

## ADAMS APPROVAL VOTING: DEFINITION

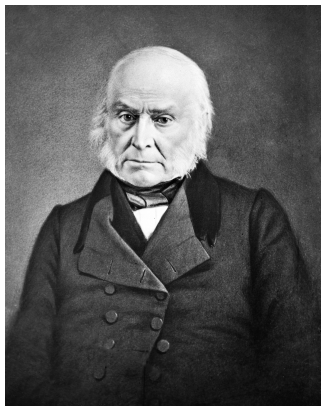
The rule is named after the sixth president of the United States, John Quincy Adams, and the electoral formula he proposed.

### Definition

Adams Approval Voting is the composite Thiele rule defined by

1.  $\omega_1(\ell) = (1, 0, 0, 0, 0, \dots)$  and
2.  $\omega_2(\ell) = (1, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots)$ .

We extend it from parliamentary elections to committee voting.



## ADAMS APPROVAL VOTING: EXPLANATION

Concretely, the rule proceeds as follows:

1. Select all committees  $W$  maximising the number of ‘covered’ voters, i.e., that maximise

$$|\{i \in N : A_i \cap W \neq \emptyset\}|$$

2. Among those, select all committees  $W'$  that maximise

$$\sum_{i \in N} \sum_{\ell=2}^{|A_i \cap W'|} \frac{1}{\ell - 1}$$

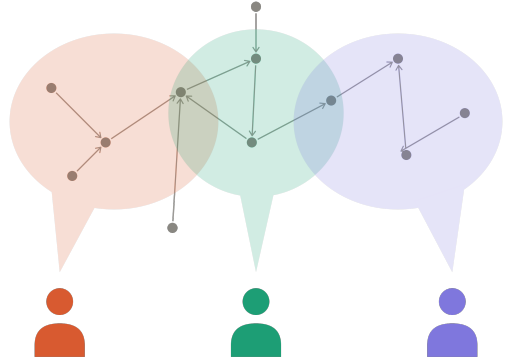
## SUPERCUT: MORE RESULTS ON REPRESENTATION

...

## 2. MANIPULATION



### 3. DELIBERATION



THANK YOU

